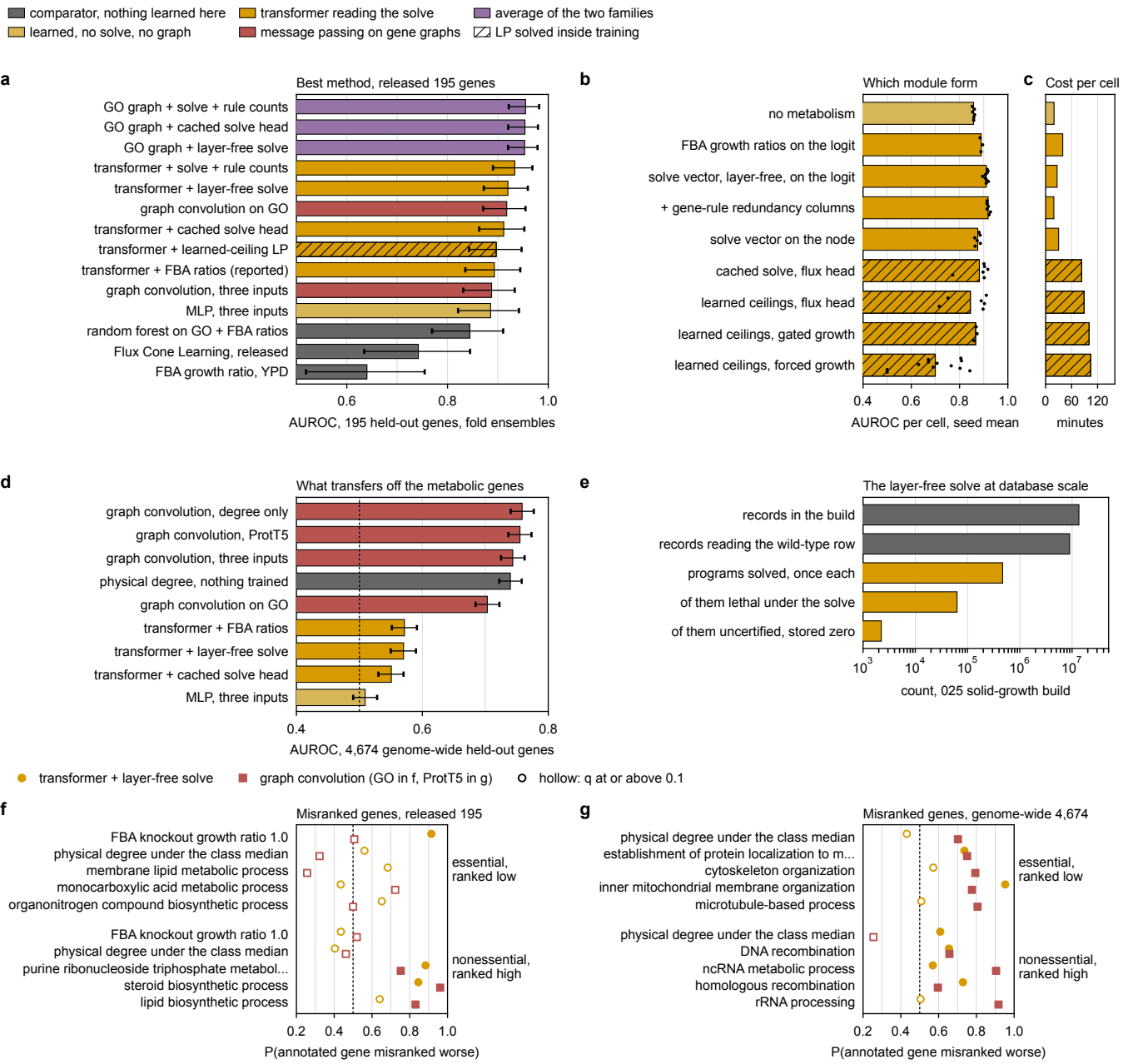

comparator, nothing learned here
learned, no solve, no graph
transformer reading the solve
message passing on gene graphs
average of the two families
LP solved inside training
a
Best method, released 195 genes
GO graph + solve + rule counts
GO graph + cached solve head
GO graph + layer-free solve
transformer + solve + rule counts
transformer + layer-free solve
graph convolution on GO
transformer + cached solve head
transformer + learned-ceiling LP
transformer + FBA ratios (reported)
graph convolution, three inputs
MLP, three inputs
random forest on GO + FBA ratios
Flux Cone Learning, released
FBA growth ratio, YPD
0.6
0.8
1.0
AUROC, 195 held-out genes, fold ensembles
b
Which module form
no metabolism
FBA growth ratios on the logit
solve vector, layer-free, on the logit
+ gene-rule redundancy columns
solve vector on the node
cached solve, flux head
learned ceilings, flux head
learned ceilings, gated growth
learned ceilings, forced growth
0.4
0.6
0.8
1.0
AUROC per cell, seed mean
c
Cost per cell
0
60
120
minutes
d
What transfers off the metabolic genes
graph convolution, degree only
graph convolution, ProtT5
graph convolution, three inputs
physical degree, nothing trained
graph convolution on GO
transformer + FBA ratios
transformer + layer-free solve
transformer + cached solve head
MLP, three inputs
0.4
0.6
0.8
AUROC, 4,674 genome-wide held-out genes
e
The layer-free solve at database scale
records in the build
records reading the wild-type row
programs solved, once each
of them lethal under the solve
of them uncertified, stored zero
10^3
10^4
10^5
10^6
10^7
count, 025 solid-growth build
transformer + layer-free solve
graph convolution (GO in f, ProtT5 in g)
hollow: q at or above 0.1
f
Misranked genes, released 195
FBA knockout growth ratio 1.0
physical degree under the class median
membrane lipid metabolic process
monocarboxylic acid metabolic process
organonitrogen compound biosynthetic process
FBA knockout growth ratio 1.0
physical degree under the class median
purine ribonucleoside triphosphate metabol...
steroid biosynthetic process
lipid biosynthetic process
essential, ranked low
nonessential, ranked high
0.2
0.4
0.6
0.8
1.0
P(annotated gene misranked worse)
g
Misranked genes, genome-wide 4,674
physical degree under the class median
establishment of protein localization to m...
cytoskeleton organization
inner mitochondrial membrane organization
microtubule-based process
physical degree under the class median
DNA recombination
ncRNA metabolic process
homologous recombination
rRNA processing
essential, ranked low
nonessential, ranked high
0.2
0.4
0.6
0.8
1.0
P(annotated gene misranked worse)